Promptable Concept Segmentation from Above: Evaluating SAM 3's Zero-Shot and One-Shot Capabilities in Remote Sensing

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***Abstract*—The deployment of large-scale foundation models, such as the Segment Anything Model 3 (SAM 3), promises a transition toward open-vocabulary, training-free computer vision. However, their capacity to generalize out-of-distribution to the complex, top-down geometric structures of Earth Observation imagery remains largely unquantified. Driven by SAM 3's performance disparities in highly specialized domains, we present a comprehensive, multi-task empirical evaluation across remote sensing scene classification, object detection, and instance segmentation under strict zero-shot and one-shot constraints. To achieve this, we introduce a structural adaptation of SAM 3 by repurposing its decoupled binary presence head into a standalone zero-shot classifier. Furthermore, by systematically isolating textual and visual prompt modalities across five configurations, we explicitly diagnose the alignment mechanics within the model's multimodal decoder. Our findings reveal severe cross-modal interference: while visual prompts successfully align the decoder to complex remote sensing geometry, textual prompts inject misaligned, ground-level semantic bias, actively degrading coordinate regression. To benchmark these capabilities without resource-intensive training, we formulate a novel training-free proxy evaluation protocol for Generalized Zero-Shot tasks (scene classification and instance segmentation). Ultimately, our results demonstrate that SAM 3 avoids the overfitting commonly seen in legacy domain-adapted models, achieving high Harmonic Mean scores in segmentation tasks. However, it remains fundamentally constrained by sub-pixel resolution limits and overhead semantic blind spots, charting a definitive mandate for parameter-efficient geospatial fine-tuning of its multimodal decoder.**

***Index Terms*—Cross-Modal Alignment, Earth Observation, Generalized Zero-Shot Learning, Scene Classification, Instance Segmentation, Open-Vocabulary Object Detection, Promptable Concept Segmentation, Remote Sensing, Segment Anything with Concepts (SAM 3), Vision-Language Models.**

I. Introduction

THE emergence of massive vision-language foundation models has transitioned the computer vision field away from dataset-specific, closed-set architectures toward open-vocabulary architectures. Central to this evolution is the concept of promptable segmentation, introduced by the Segment Anything Model (SAM) [1]. SAM shifted from class-specific segmentation to class-agnostic segmentation guided by spatial prompts, a capability subsequently extended to temporal video tracking by SAM 2 [2]. The deployment of SAM 3 [3] advances this class-agnostic foundation toward a unified task defined as Promptable Concept Segmentation (PCS). By integrating a multimodal decoder and a decoupled presence head, SAM 3 separates global semantic recognition from localized spatial bounding. This decoupled architecture unifies zero-shot conceptual reasoning with precise mask generation. However, initial benchmark data reveal a critical limitation. While SAM 3 achieves 56.4 AP on general-domain datasets like COCO, its zero-shot performance drops to 15.2 AP on the RF-100VL benchmark [3], which comprises 100 specialized visual domains. This performance gap indicates that while SAM 3 acts as an effective generalist, its direct application to highly specialized fields, such as Earth Observation (EO) and Remote Sensing (RS), requires objective evaluation to determine if costly domain-specific fine-tuning is necessary.

The direct application of general-domain foundation models to remote sensing imagery is obstructed by structural, geometric, and semantic domain gaps. Unlike standard consumer photography, satellite and aerial imagery are characterized by a nadir (top-down) perspective, multi-scale distributions, dense background clutter, and an absence of canonical orientation [4]. This spatial anisotropy induces a cross-modal alignment failure within the model's multimodal fusion layers. Because standard vision-language models are optimized contrastively over ground-level image-caption pairs, their text encoders map concepts to latent vectors based on horizontal viewpoints. For example, the linguistic embedding of a "school" is implicitly tied to its vertical facades and windows. However, when viewed from an overhead satellite perspective, these objects collapse into flat, two-dimensional geometric patches. Consequently, the multimodal decoder fails to reconcile ground-level textual expectations with top-down visual features. This creates semantic blind spots, despite the model possessing a robust spatial feature extraction backbone.

To address this cross-modal alignment deficit, existing remote sensing frameworks undergo supervised fine-tuning over a limited set of base aerial categories. While this supervised calibration successfully encodes dataset-specific contextual and spatial priors, it simultaneously introduces overfitting and semantic bias [5]. Because the projection layers overfit to the training distribution, the network learns to either misclassify novel targets as familiar base classes or suppress them entirely as background noise. Consequently, upon encountering novel, unseen classes during inference, the performance of these specialized Generalized Zero-Shot Detection (GZSD) and Instance Segmentation (GZSI) models significantly degrades [5], [6].

To resolve this ambiguity and establish a performance baseline, this study evaluates SAM 3 under zero fine-tuning.

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We analyze the multimodal processing of the model across the AID [7], DIOR [8], and iSAID [9] datasets by varying the input prompts. These benchmarks serve as the established standards for evaluation in the remote sensing literature. We adapt SAM 3's decoupled architecture by utilizing its binary presence head as a standalone global scene classifier. Furthermore, we define a matrix of five prompt configurations (Configurations **Conf 1** through **Conf 5**) to map how SAM 3's multimodal decoder processes remote sensing input prompts. These configurations isolate textual prompts, visual prompts, multimodal prompt combinations, and oracle negative filtering. Finally, to bridge the evaluation gap between frozen foundation models and supervised models, we formulate a novel proxy evaluation protocol. We map the one-shot multimodal prompt configurations (Configurations **Conf 3** or **Conf 5**) to the community's standard "Base" class split and the zero-shot textual prompt configuration (Configurations **Conf 1** or **Conf 4**) to the "Novel" class split. This mirrors standard Generalized Zero-Shot evaluation protocols, allowing us to compute the Harmonic Mean and benchmark against existing models while remaining entirely training-free.

The primary contributions of this work are summarized as follows:

- We introduce a structural adaptation of SAM 3, utilising its decoupled binary presence head as a standalone zero-shot scene classifier to evaluate the global semantic awareness of its multimodal decoder.
- We conduct a prompt ablation study using five configurations across object detection and instance segmentation. This isolates the cross-modal interference within the multimodal decoder, demonstrating that visual prompts achieve spatial alignment while textual prompts introduce perspective bias.
- We formulate a novel, training-free proxy evaluation methodology that maps the visual and textual prompt configurations of frozen foundation models directly to standard Generalized Zero-Shot (GZSD/GZSI) conditions. This establishes a standardised baseline without requiring closed-set fine-tuning.
- We provide a class-by-class diagnostic analysis detailing how target size, visual complexity, and geometric invariance influence the performance of zero-shot geospatial foundation models. This analysis identifies a resolution bottleneck regarding small-scale feature representation.

The remainder of this paper is organized as follows. Section II reviews the literature on vision-language models, SAM promptable segmentation, and generalized zero-shot learning for remote sensing. Section III describes the benchmark datasets, the proposed adaptation of SAM 3 for zero-shot scene classification, the prompting configurations, and the evaluation protocol. Section IV presents the experimental results on scene classification, object detection, and instance segmentation, together with quantitative and qualitative analyses. Finally, Section V summarizes the main findings and outlines directions for future research.

II. Related Work

A. Vision-Language Models and Zero-Shot Scene Classification

The development of Vision-Language Models (VLMs), such as OpenAI CLIP [10], established the foundation for open-vocabulary scene understanding through large-scale contrastive learning. By optimizing a bi-directional InfoNCE loss, these models map visual and textual feature vectors into a shared latent embedding space. While standard CLIP demonstrates robust scene comprehension on ground-level photography, its direct application to Earth Observation (EO) imagery exposes a semantic domain gap [4]. Satellite imagery is characterized by a nadir (top-down) perspective, dense multi-scale object distributions, and high inter-class similarity. For example, distinguishing between "medium residential" and "dense residential" zones requires fine-grained feature extraction. Consequently, the visual feature vectors extracted from overhead perspectives are shifted in the latent space, failing to align with their corresponding textual feature vectors [11].

To address this perspective mismatch, recent models employ domain-specific continuous pre-training. Architectures such as RemoteCLIP [4], GeoRSCLIP [11], and RSDiX [12] adapt foundation models by fine-tuning on large datasets using rule-based annotations or self-distillation techniques. While this targeted contrastive alignment improves whole-scene aerial classification accuracy, these VLMs are constrained by their global image-level objectives. Consequently, they lack the dense prediction capabilities required for bounding box localization and precise mask generation.

B. The Evolution of Promptable Architectures: From SAM to SAM 3

To decouple spatial localization from closed-set classification, the original Segment Anything Model (SAM) [1] introduced promptable visual segmentation. It utilized a Vision Transformer (ViT) image encoder paired with a prompt-driven mask decoder. However, its core design was class-agnostic. While it achieved zero-shot localization guided by spatial prompts (points or boxes), it lacked a semantic latent space to comprehend and segment based on open-vocabulary textual concepts. Its successor, SAM 2 [2], extended this promptable architecture into the temporal domain to enable continuous video object tracking. Yet, despite its temporal advancements, it operated primarily as a class-agnostic localizer lacking the explicit multimodal alignment required for zero-shot semantic recognition.

SAM 3 [3] introduces an architecture designed to address Promptable Concept Segmentation (PCS). As illustrated in Figure 1, SAM 3 transitions from a purely spatial system to a unified multimodal framework. It expands the promptable interface by introducing a dedicated Text Encoder and an Exemplar Encoder to process text and visual prompts, respectively. These extracted prompt features, alongside the base Image Features, are fed into the Multimodal Decoder to generate Conditioned Image Features. The architecture then bifurcates. A Pixel Decoder upsamples the Conditioned Image Features to generate high-resolution per-pixel embeddings.

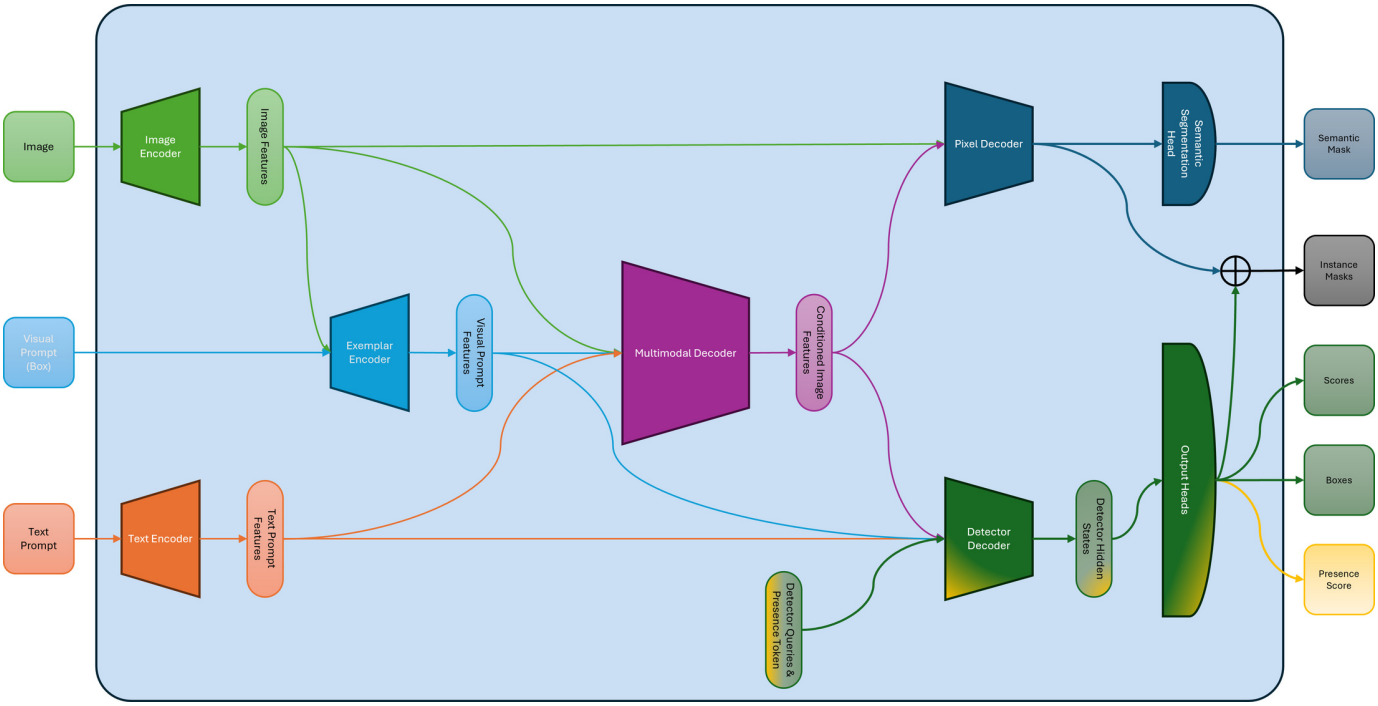


Fig. 1: The architectural evolution of SAM 3, highlighting the integration of the Multimodal Decoder, specialized prompt encoders, and the decoupled Presence Head used to achieve Promptable Concept Segmentation [3].

These embeddings are routed to a Semantic Segmentation Head to produce a broad Semantic Mask. Simultaneously, a Detector Decoder processes the Conditioned Image Features, the original prompt features, and a fixed set of learned Detector Queries paired with a Presence Token. This design is heavily inspired by the DEtection TRansformer (DETR) paradigm [13], where a fixed number of learned object queries interact with encoded features via cross-attention. Through these mechanisms, the Output Heads translate the decoded states into localized bounding boxes and final semantic scores. Crucially, to generate precise Instance Masks, the pixel-level embeddings from the Pixel Decoder are mathematically fused with the object-level queries from the Output Heads.

A fundamental challenge in open-vocabulary segmentation is the architectural trade-off between semantic recognition and spatial localization. Aggregating global context to resolve semantic ambiguity often requires feature pooling, which degrades the fine-grained spatial details required for precise mask boundaries.

To resolve this architectural trade-off, SAM 3 introduces a decoupled Presence Head. Instead of requiring the local detector queries to independently predict both the global existence of a concept and its exact boundaries, the architecture decomposes the prediction. A shared, learned Presence Token independently evaluates the Global Presence Score, which represents the probability that the requested concept exists anywhere in the scene. The final match probability for each individual query is factored as:

$$p(\text{query}_i \text{ matches NP}) = p(\text{query}_i \text{ matches NP} \mid \text{NP appears}) \cdot p(\text{NP appears}) \quad (1)$$

Conditioned on this global presence, the local object queries can focus strictly on foreground-background localization without being constrained by global contextual demands. This decoupled approach prevents localized queries from being overwhelmed by global classification tasks, thereby improving detection precision and reducing false positives in dense environments.

C. Generalized Zero-Shot Detection (GZSD) and Instance Segmentation (GZSI)

To identify and segment novel classes without continuous retraining, the remote sensing community investigates Generalized Zero-Shot Detection (GZSD) and Instance Segmentation (GZSI). The GZSD trajectory has evolved from generative feature synthesis to deep multimodal alignment. Early generative frameworks, such as RRFS [14], converted zero-shot tasks into supervised problems by synthesizing visual features for unseen categories. These models utilized Wasserstein GANs (WGAN) and intra-class semantic diverging techniques to manage the high intra-class diversity and inter-class separability inherent to aerial targets. Subsequent architectures like Grounding DINO [15] and CoseDet [16] shifted toward dense cross-attention mechanisms. CoseDet, for example, replaces standard CLIP backbones with RemoteCLIP and applies pseudo-word tokens to leverage regional scene context to disambiguate dense aerial targets.

In the GZSI domain, models predict pixel-level masks rather than bounding boxes. Early models such as ZSI [6] and D2Zero [5] focused on mitigating background ambiguity and semantic bias during novel class inference through semantic-promoted debiasing. More recently, the ZoRI architecture [17]

addresses performance degradation on aerial imagery by refining frozen CLIP embeddings via a Discriminative-Enhanced Classifier. Furthermore, it employs Knowledge-Maintained Adaptation to learn domain-specific textures without disrupting the pre-trained vision-language alignments.

Despite these methodological advancements, contemporary GZSD and GZSI architectures rely on supervised fine-tuning over a restricted subset of "Seen" aerial base categories [5], [17]. This supervised calibration phase binds the projection layers to the training distribution, biasing the model toward seen classes and degrading generalization when encountering "Unseen" novel targets.

This study bridges the evaluation gap between frozen foundation models and these supervised architectures. Because we evaluate SAM 3 without fine-tuning, traditional training on Base class splits is not applicable. To address this, we introduce a training-free proxy evaluation. We map SAM 3's multimodal prompt configurations (Configurations **Conf 3** and **Conf 5**) to the "Base" performance, and its textual prompt configurations (Configurations **Conf 1** and **Conf 4**) to "Novel" zero-shot performance. This proxy allows us to compute the Harmonic Mean, establishing a standardized benchmark against existing GZSD/GZSI models. Ultimately, this framework demonstrates that frozen foundation models can maintain an open vocabulary and avoid the overfitting associated with supervised fine-tuning.

III. Methodology

A. Datasets

To evaluate the zero-shot and one-shot capabilities of SAM 3 across diverse remote sensing tasks, we selected three distinct benchmark datasets, covering classification, detection, and segmentation:

- The **Aerial Image Dataset (AID)** [7] is a benchmark for aerial scene classification, comprising 10,000 multi-source Google Earth images distributed across 30 semantic scene categories, including airport, farmland, industrial, and residential areas. Each image has a fixed spatial dimension of 600×600 pixels, with spatial resolution ranging from 0.5 m to 8 m per pixel depending on the source imagery. The dataset is partitioned into training and test subsets at a standard 50%/50% ratio. Key challenges include substantial intra-class variation and inter-class similarity arising from the multi-source nature of the imagery.
- The **Object Detection in Optical Remote Sensing Images Dataset (DIOR)** [8] is a benchmark for object detection in optical remote sensing imagery, comprising 23,463 images of 800×800 pixels with 192,472 annotated instances across 20 object categories labeled with horizontal bounding boxes. Spatial resolution ranges from 0.5 m to 30 m per pixel, with images sourced from over 80 countries. The dataset is partitioned into training, validation, and test subsets of 5,862, 5,863, and 11,738 images, respectively. Key challenges include wide object-scale variation, high inter-class similarity, and significant diversity in imaging conditions across weather, seasons, and viewpoints.
- The **Instance Segmentation in Aerial Images Dataset (iSAID)** [9] is a benchmark for instance segmentation in aerial imagery, comprising 2,806 high-resolution images with 655,451 densely annotated object instances across 15 categories, combining instance-level detection with pixel-level segmentation masks. Image sizes range from 800×800 to $4,000 \times 13,000$ pixels, collected from multiple sensors and platforms at varying spatial resolutions. The dataset is partitioned into training, validation, and test subsets of 1,411, 458, and 937 images, respectively. Key challenges include large object-scale variation, high instance density per image, and the prevalence of small objects.

B. Adapting SAM 3 for Zero-Shot Scene Classification

Because SAM 3 is a promptable binary segmentation model designed for localized mask generation, it requires structural adaptation for whole-image scene classification. We utilise its decoupled architecture, specifically the independent Presence Head, to evaluate classification in a zero-shot setting.

Let $\mathbf{I}$ represent an input image and $\mathcal{C} = \{c_1, c_2, \ldots, c_N\}$ denote the set of N target concepts in a given dataset (where $N = 30$ for the AID dataset). For each concept $c \in \mathcal{C}$, we independently query SAM 3 to yield two scalar outputs:

1) **Presence Score (P_c):** The global token probability indicating the existence of concept c within $\mathbf{I}$.
2) **Conditional Localisation Score (L_c):** The highest localisation score assigned by the Output Heads, conditioned on the presence of the concept.

To establish the final predicted image class $\hat{y}$, we formally evaluate three decision functions. The baseline functions rely on standalone probabilities:

$$\hat{y}_{\text{pres}} = \arg\max_{c \in \mathcal{C}} P_c \tag{2}$$

$$\hat{y}_{\text{local}} = \arg\max_{c \in \mathcal{C}} L_c \tag{3}$$

To fuse global semantic context with localized geometric confidence, we compute a combined Confidence Score (S_c). This formulation natively reconstructs SAM 3's decoupled query match probability (defined in Equation 1) by multiplying the global presence probability by the local conditional confidence:

$$S_c = P_c \cdot L_c \tag{4}$$

This yields the final decision function:

$$\hat{y}_{\text{combined}} = \arg\max_{c \in \mathcal{C}} S_c \tag{5}$$

C. Formalizing Prompt Modalities and Negative Filtering

For open-vocabulary object detection and instance segmentation, we evaluate the model by prompting each target class independently per image. To systematically isolate the impact of different prompt modalities, we define the input

prompt set $\mathcal{Q}_c$ for a target class c. Let t_c denote the textual prompt (represented as a short noun phrase of the class name) and b_c denote the visual prompt (provided as a ground-truth visual exemplar cropped from the object's bounding box). We establish three core experimental modalities:

Conf 1 - **Text-Only:** Zero-shot semantic inference, where $\mathcal{Q}_c = \{t_c\}$.

Conf 2 - **Box-Only:** Spatial inference bypassing the text encoder, where $\mathcal{Q}_c = \{b_c\}$.

Conf 3 - **Text + Box:** Multimodal one-shot inference combining both prompt types, where $\mathcal{Q}_c = \{t_c, b_c\}$.

To evaluate the limit of the decoupled presence head and track the impact of false-positive noise, we introduce an oracle ground-truth negative filtering. Let $f_c \in \{0, 1\}$ be a binary indicator function that returns 1 if at least one instance of concept c is present in the ground truth of image **I**, and 0 otherwise. The oracle Confidence Score $\tilde{S}_c$ is computed as:

$$\tilde{S}_c = S_c \cdot f_c \tag{6}$$

This formulation suppresses the model's output by forcing the confidence score to absolute zero for any category not represented in the image background. This negative suppression mechanism yields two additional evaluation configurations:

Conf 4 - **Text + Oracle (filtered) :** Configuration **Conf 1** subjected to oracle negative suppression.

Conf 5 - **Text + Box + Oracle (filtered):** Configuration **Conf 3** subjected to oracle negative suppression.

Comparing the filtered configurations' output against the unfiltered baselines quantifies the model's susceptibility to false-positive when prompted with absent concepts.

D. Evaluation Metrics and Proxy Generalization Framework

Performance for both object detection and instance segmentation is quantified using the mean Average Precision (mAP) averaged across Intersection over Union (IoU) thresholds from 0.50 to 0.95, as well as the mAP at the 0.50 threshold (AP_{50}).

Standard Generalized Zero-Shot (GZSD/GZSI) models in the literature are evaluated by splitting dataset classes into a "Base" distribution (seen during training) and a "Novel" distribution (unseen during training), as detailed in Section II-C. Because our objective is to benchmark SAM 3 without supervised fine-tuning, evaluating across a traditional supervised split is inapplicable.

To enable direct comparison against these specialized models, which predominantly rely on the AP_{50} metric for benchmarking, we employ a proxy calculation that mirrors their visual evaluation constraints:

- **Base AP_{50} (AP^B_{50}):** Extracted from the Text + Box configuration (Configuration **Conf 3**). This visual box exemplar serves as a proxy for the spatial and structural priors a traditionally trained model possesses on familiar base classes.
- **Novel AP_{50} (AP^N_{50}):** Extracted from the Text-Only configuration (Configuration **Conf 1**). This constrains the model to rely solely on language-to-vision alignment, acting as a proxy for the cross-modal semantic inference that traditional models perform on novel categories.

To measure open-vocabulary generalizability without penalizing balanced model, we compute the standard Harmonic Mean (HM) at the 50% IoU threshold using these baseline metrics:

$$\text{HM} = \frac{2 \cdot \text{AP}^B_{50} \cdot \text{AP}^N_{50}}{\text{AP}^B_{50} + \text{AP}^N_{50}} \tag{7}$$

To evaluate performance under the oracle setting, we substitute the baseline configurations with their negative-filtered counterparts. Specifically, we use Configuration **Conf 5** to obtain the oracle base performance ($\tilde{\text{AP}}^B_{50}$) and Configuration **Conf 4** to obtain the oracle novel performance ($\tilde{\text{AP}}^N_{50}$). This yields the Oracle Harmonic Mean ($\tilde{\text{HM}}$):

$$\tilde{\text{HM}} = \frac{2 \cdot \tilde{\text{AP}}^B_{50} \cdot \tilde{\text{AP}}^N_{50}}{\tilde{\text{AP}}^B_{50} + \tilde{\text{AP}}^N_{50}} \tag{8}$$

This formulation establishes a standardized mathematical metric to contrast SAM 3's generalized zero-shot and one-shot capabilities directly against domain-adapted models.

IV. Experimental Results and Discussion

A. Aerial Scene Classification (AID)

We evaluate SAM 3 on the 30-class AID dataset in a zero-shot setup as explained in III-B. The macro performance results are consolidated alongside a cross-model baseline evaluation in Table I. The baseline performance metrics for OpenAI CLIP L/14 are reported as reproduced by SkyScript [18] while the metrics for the B/32 and Resnet-50 by RemoteCLIP [4]. The complete, 30-class performance target matrix is provided in Table II. Corresponding qualitative results are illustrated in Figure 2.

1) Scoring Head Dynamics (Presence vs. Localisation): As shown in Table I, the classification strategy guided by the standalone Presence Score (Equation 2) achieves an accuracy of 37.26%, outperforming the Conditional Localisation Score (Equation 3), which yields 28.25%. This performance gap highlights the structural decoupling of the model. The independent presence head evaluates the global existence of the target concept across the entire scene embedding, whereas the output heads verify the presence of the specific conceptual prompt within localized sub-regions of the shared space, conditioned on the concept existing in the image.

Fusing these outputs via the combined Confidence Score (Equation 5) yields the highest classification accuracy of 38.70%, which indicates that these decoupled predictions provide complementary representations. The Localisation score provides conditional localized verification of the concept within the shared embedding space, while the global presence score provides a macro-level prior for the overall scene context. Consequently, this joint decision rule enforces a joint probability constraint that requires a concept to satisfy both global and localized verification criteria simultaneously, improving scene classification performance compared to either standalone evaluation score.

2) Granular Per-Class Divergence: Vocabulary vs. Geometry: To understand the mechanics of SAM 3's zero-shot classification, we analyse the quantitative class-wise performance detailed in Table II alongside the qualitative visual evidence

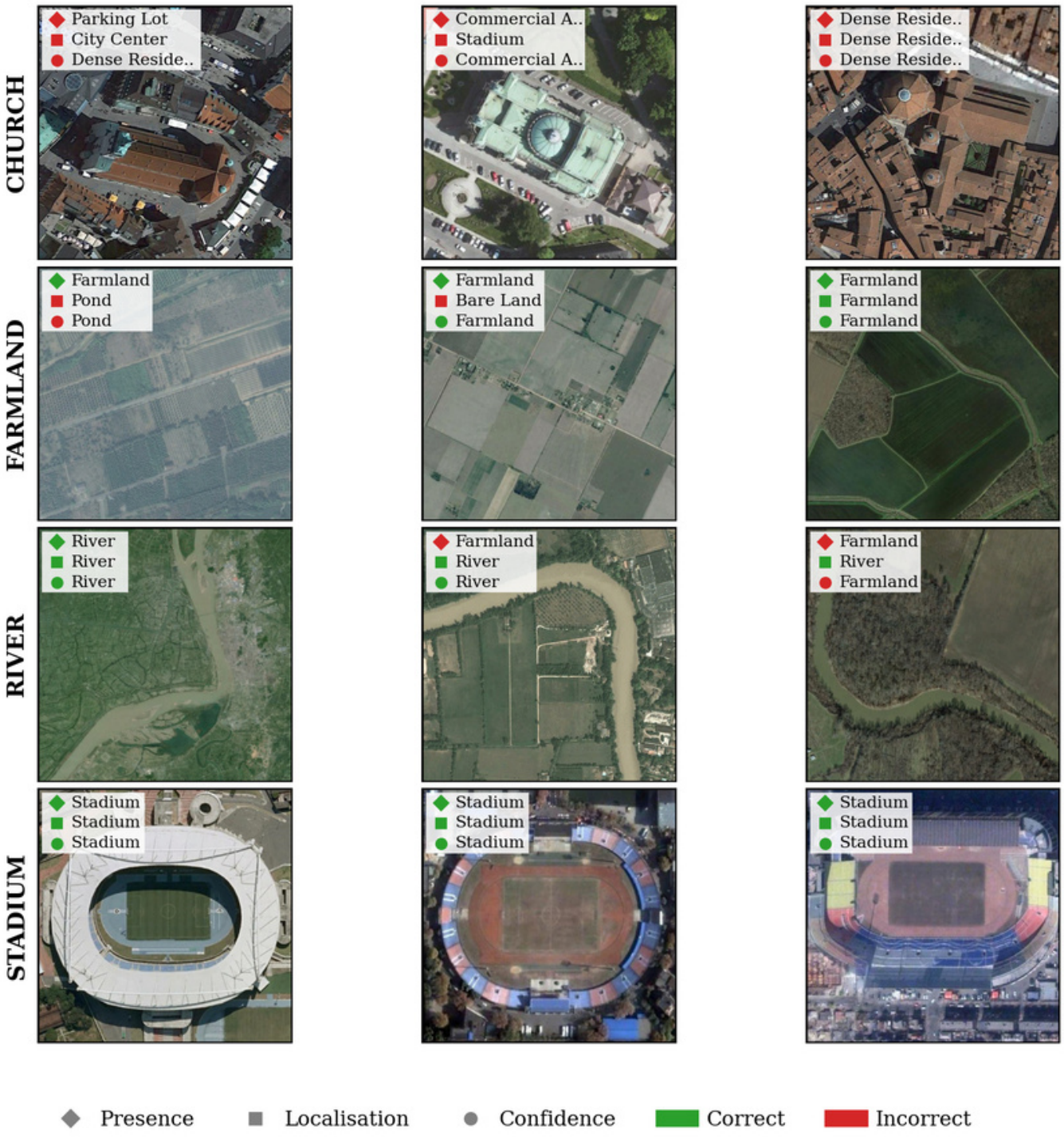


Fig. 2: Qualitative zero-shot scene classification results on selected categories from the AID dataset. Each row displays three distinct image examples for a target class, illustrating different combinations of successful and unsuccessful predictions. Marker shapes correspond to the three evaluated decision functions: the Presence Score ($\hat{y}_{\text{pres}}$), indicated by a diamond ($\diamond$); the Conditional Localisation Score ($\hat{y}_{\text{local}}$), indicated by a square ($\square$); and the Confidence Score ($\hat{y}_{\text{combined}}$), indicated by a circle ($\circ$). Green markers denote correct classifications, while red markers denote incorrect classifications. The rows demonstrate four distinct outcomes: (1) *Church*, where all three decision functions produce incorrect classifications across the provided examples; (2) *Farmland*, where only the decision function based on the Presence Score produces correct classifications; (3) *River*, where only the decision function based on the Conditional Localisation Score produces correct classifications; and (4) *Stadium*, where all three decision functions produce correct classifications across all examples.

TABLE I: Cross-Model Benchmark Comparison on the AID Dataset. We report zero-shot Top-1 scene classification accuracies for SAM 3 and compare them against existing foundational and remote-sensing-adapted vision-language models. Baseline results for all competing methods are taken directly from their respective original publications or reported benchmark studies. SAM 3 is evaluated using three decision strategies derived from its promptable segmentation architecture: Presence Score, Conditional Localisation Score, and their multiplicative Confidence Score. All methods correspond to zero-shot settings on the AID dataset.

Model Family	Specific Variant	Accuracy (%)
OpenAI CLIP [10]	ResNet-50	57.35
OpenAI CLIP [10]	B/32	65.65
OpenAI CLIP [10]	L/14	69.25
RemoteCLIP [4]	ResNet-50	86.55
RemoteCLIP [4]	B/32	91.30
RemoteCLIP [4]	L/14	87.90
SkyCLIP-50 [18]	B/32	70.90
SkyCLIP-50 [18]	L/14	71.70
RS-CLIP [19]	B/32	65.48
GeoRSCLIP [11]	B/32	73.72
GeoRSCLIP [11]	H/14	76.33
RSDiX-CLIP [12]	B/32	**95.10**
RSDiX-CLIP [12]	B/16	92.60
RSDiX-CLIP [12]	L/14	91.10
HQRS-CLIP [20]	-	73.86
SAM 3	Presence Score (2)	37.26
SAM 3	Conditional Localization Score (3)	28.25
SAM 3	Confidence Score (4)	**38.70**

TABLE II: Per-Class Scoring and Sample Breakdown on the AID Benchmark Dataset. The metrics represent the zero-shot Top-1 classification accuracy achieved under each scoring criterion.

Class Concept	Presence Accuracy (%)	Localisation Accuracy (%)	Confidence Accuracy (%)
church	0	7.92	0
stadium	80.34	55.52	90.69
railway station	0.38	0.77	0.77
farmland	77.03	10.00	65.95
baseball field	77.73	50.45	79.55
storage tanks	78.61	39.17	76.39
beach	94.00	28.00	93.50
parking lot	100	32.05	93.33
airport	16.39	17.78	21.39
forest	70.40	80.40	72.40
park	0.57	0.57	0.57
river	44.15	87.07	59.51
school	0	0	0
sparse residential	0	0	0.33
pond	44.76	81.19	57.14
bare land	81.61	74.84	86.13
dense residential	45.12	19.76	54.39
playground	1.08	0.27	0.54
medium residential	50.00	30.00	50.69
desert	35.67	60.67	37.00
city center	0	1.15	0
resort	9.66	0.69	4.14
industrial	1.28	4.36	4.36
public square	0.91	3.03	0.61
viaduct	0	0	0
mountain	67.35	65.29	70.00
bridge	63.61	46.11	66.11
port	0.26	0	0.26
meadow	33.21	48.57	35.00
commercial area	28.29	3.71	20.57

presented in Figure 2. If we conceptually map this data, the target classes naturally partition into distinct operational clusters. These groupings illustrate how the inherent physical traits of a class govern which decision function ultimately succeeds or fails:

- **Global Scene Dominance (Presence Wins):** When an entire scene inherently embodies a specific concept without relying on a single, isolated geometric boundary, the global presence token dominates the classification. For example, *farmland* yields a strong Presence Score (77.03%) but a poor Conditional Localisation Score (10.00%). The *farmland* row in Figure 2 demonstrates this perfectly: the Presence Score accurately captures the global scene context, whereas the Conditional Localisation Score gets distracted by localized sub-features, mistakenly highlighting patches of bare land or ponds.
- **Distinct Feature Geometry (Localisation Wins):** Conversely, when a target class is defined by large, clear, and easily isolatable physical features, localized geometric confidence drives the prediction. Natural amorphous features like *rivers* yield a high Conditional Localisation Score (87.07%) but a much lower Presence Score (44.15%). As seen in the *river* row of Figure 2, the Conditional Localisation Score accurately traces the land-water boundaries to classify the image, while the global Presence Score misclassifies the broader surrounding scene as farmland.
- **Synergistic Success (Both Win):** When a concept possesses both a massive, unambiguous geometric footprint and overwhelmingly dictates the global scene context, both heads succeed. The model performs exceptionally well on structures like *parking lots* (100% Presence, 93.33% Confidence) and *stadiums* (80.34% Presence, 90.69% Confidence). This synergy is visually confirmed in the *stadium* examples in Figure 2, where all three decision functions unanimously and correctly identify the target.
- **Cross-Modal Domain Failure (Both Fail):** Because SAM 3 is trained on a general-domain distribution, it severely struggles with concepts where it can neither reconcile the global scene nor isolate the defining local features from an overhead perspective. Accuracy falls to near zero for human-designated architectural properties like *schools*, *churches*, and *city centers*. The *church* row in Figure 2 illustrates this complete semantic collapse, with all three decision functions failing entirely and hallucinating unrelated classes. This is not a failure of understanding the textual concept itself, but a profound cross-modal alignment gap. While a church appears as basic geometric rectangles from a nadir satellite view, the model's text-to-vision alignment relies on ground-level indicators (e.g., steeples, facades). Consequently, the top-down visual embeddings fail to align with the pre-trained text embeddings, causing both the global and local scoring heads to fail completely.

3) Cross-Model Comparison and Fine-Tuning Potential: Comparing these results with other vision-language models in

Table I provides context regarding SAM 3's current limitations and downstream potential. The baseline accuracy for the OpenAI CLIP L/14 model is reported as benchmarked by Wang et al. [18]. The general-domain OpenAI CLIP [10] outpaces SAM 3 with a zero-shot accuracy of 69.25%, as CLIP was engineered for global image-text contrastive learning to map holistic scene features directly to a text lexicon. Furthermore, when CLIP is fine-tuned on earth observation datasets, such as in GeoRSCLIP [11] and RSDiX-CLIP [12], its scene classification accuracy reaches 73.72% and 95.10%, respectively.

This comparison illustrates both the current limitations and the potential of SAM 3. Currently, SAM 3 is constrained by its zero-shot vocabulary mapping, which aligns with physical ground-level object shapes rather than global aerial zoning. However, the class-level performance variations highlighted in the per-class analysis indicate that the model's visual segmentation infrastructure can successfully isolate aerial masks without prior domain training. Because the underlying spatial backbone is highly effective, a lightweight parameter-efficient fine-tuning stage or a domain-specific text projection layer could address these semantic misalignments, thereby enhancing performance in remote sensing applications.

B. Open-Vocabulary Object Detection (DIOR)

We evaluate SAM 3 within an open-vocabulary detection framework using the 20-class DIOR dataset. The performance metrics across our five core prompting configurations are summarized in Table IV. To observe the localized feature interactions underlying these scores, a qualitative prompt ablation across four distinct environments is presented in Figure 3.

1) Cross-Modal Alignment Gap and Linguistic Interference: Analyzing the experimental configuration reveals a severe polarization between SAM 3's textual alignment and its pure visual parsing abilities. The zero-shot baseline using text prompts (**Conf 1**) yields an 11.67 mAP. Conversely, introducing visual guidance through a single bounding box exemplar (**Conf 2**) forces an absolute localization surge to 66.55 mAP (79.51 AP@50). This proves conclusively that SAM 3's foundational visual engine easily parses complex, dense remote sensing geometry when performing intra-modal (visual-to-visual) matching.

Crucially, fusing both modalities (Configuration **Conf 3**) compresses aggregate localization to 63.79 mAP, actively underperforming the standalone visual exemplar by 2.76 percentage points. This negative cross-modal interference is clearly illustrated by the overpass detection task in Figure 3c. Observing the image, the pure visual prompt fails to isolate the overpass, instead scattering bounding boxes across the connected highway system, while the pure text prompt yields zero detections. However, when both prompts are mixed, the model accurately isolates the specific boundaries of the overpass.

Mechanistically, this highlights how text can act as a crucial helper to shift misaligned vectors. In the shared latent space, the visual feature vector of an overpass is nearly identical to a standard road, causing the visual-only mode to over-segment. At the same time, the textual feature vector for "overpass" is closely aligned with ground-level perspectives, leaving it far away from the top-down highway features. When both prompts are fused, the textual vector pulls the combined representation away from the generic road concept and anchors it down into the correct top-down overpass layout, correcting the final bounding boxes.

2) Presence Head Filtering Analysis: Transitioning to the negative-filtered configurations provides insight into the characteristics of the presence head. Moving from **Conf 1** (11.67 mAP) to **Conf 4** (13.11 mAP) and from **Conf 3** (63.79 mAP) to **Conf 5** (64.61 mAP) yields a consistent but modest performance increase. This statistical shift indicates that without active prompting suppression, the model experiences false-positive detections over structurally similar elements.

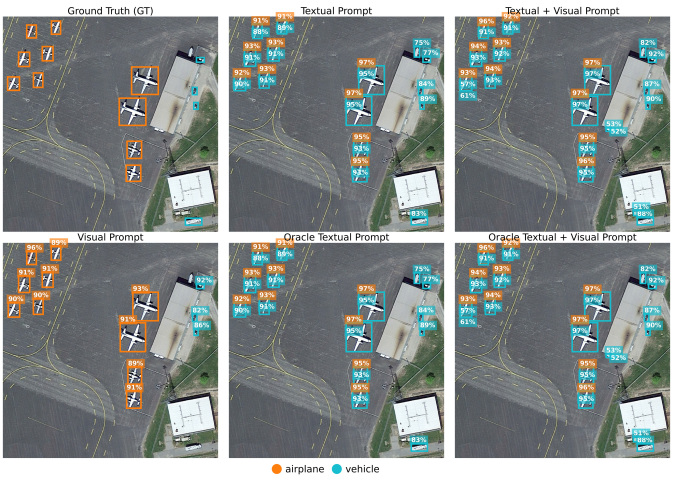
This behavior is illustrated in the windmill field example in Figure 3d. In the unfiltered text and multimodal configurations, the model misclassifies the windmills, generating false-positive "airplane" bounding boxes around them. When the negative oracle filter is applied, these false-positive airplane boxes are suppressed.

In the shared feature space, the top-down visual shape of a windmill's blades shares a structural similarity with the wings and fuselage of an aircraft. Because the text-vision alignment layers were pre-trained predominantly on ground-level imagery, the multimodal decoder struggles to map the purely semantic text embedding of "windmill" to this unfamiliar top-down visual representation. Consequently, the decoder shifts the target representation toward the "airplane" feature space, where the visual similarity is more pronounced. The oracle filter counteracts this by removing the mismatched airplane queries from the downstream decoders. This indicates that while the decoupled presence head introduces background noise, it is not the primary performance bottleneck. The main operational constraint remains false negatives driven by the difficulty of mapping semantic text prompts to top-down aerial geometries.

3) Granular Class Analysis: Persistent Vulnerabilities and Structural Drops: A class-by-class analysis of the performance matrix in Table III and the visualisation in Figure 3 reveals localized variations that persist across the five evaluation configurations:

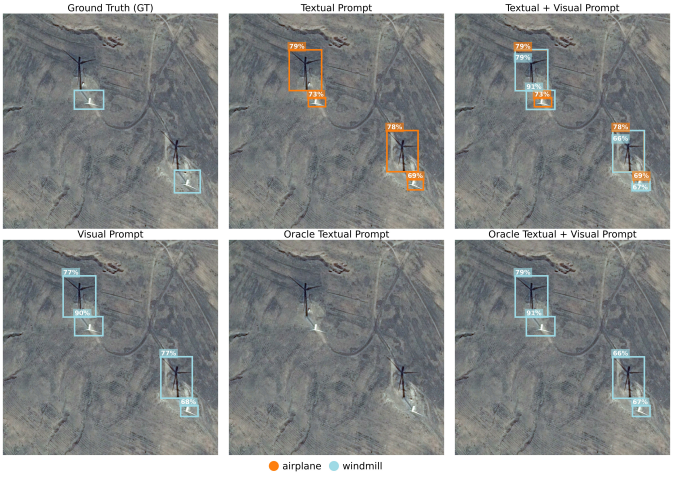
- **Semantic Performance on Macro Zones:** Large-scale spatial categories completely collapse under the pure zero-shot text modality (Configuration **Conf 1**). *Airport* drops to an absolute 0 mAP in both the Text-Only (Configuration **Conf 1**) and Text + Oracle (Configuration **Conf 4**) configurations, but immediately shoots up to 92.34 mAP under the visual prompt configuration (Configuration **Conf 2**) and 92.39 mAP under the multimodal configuration with oracle filtering (Configuration **Conf 5**). Similarly, *Baseball Field* sits at a non-functional 0.01 mAP under the pure text modality, peaks at 77.54 mAP under the visual prompt configuration, and suffers a massive drop back down to 60.98 mAP under the multimodal configuration with oracle filtering. Mechanistically, these sprawling, heterogeneous regions lack a distinct, localized visual feature vector for text

(a) Isolated Airfield



(b) Dense Complex Airfield

(c) Highway Overpass Intersection



(d) Windmill Power Generation Field

Fig. 3: Qualitative object detection results on the DIOR dataset across the five prompt configurations (Configurations **Conf 1** through **Conf 5**).)

embeddings to anchor onto. When textual and visual prompts are fused (Configuration **Conf 3**), the linguistic embedding acts as semantic noise, pulling the clean visual representation away from its target and causing the sharp drop observed in categories like the baseball field.

- **Text-Induced Clutter and Vehicle Interaction:** Unlike large zones, distinct geometric targets show strong initial text affinity, with *Airplane* scoring a high 60.22 mAP under the pure zero-shot text modality (Configuration **Conf 1**). However, unguided text embeddings trigger heavy background clutter. Observing the airplane task in Figure 3a, the text prompt successfully captures the airplanes but hallucinates false-positive boxes over background vehicles that are not labeled in the ground truth. Applying the oracle filtering (Configuration **Conf 4**) suppresses these errors. Conversely, in Figure 3b, where vehicles are actively present in the ground truth, the oracle filtering correctly leaves these vehicle boxes intact. This aligns with Table III, where *Vehicle* accuracy rises from 17.55 mAP under the Text-Only configuration to 21.47 mAP under the Text + Oracle configuration, proving that text vectors map successfully to distinct shapes but require visual prompting to prevent background spillover.
- **The Shadow Distortion Constraint:** A critical physical bottleneck is the model's inability to decouple an object's structural boundary from its projected shadow, visible across both the windmill blades in Figure 3d and the vehicles in Figure 3b. This constraint is mathematically reflected in Table III, where *Windmill* stalls at a low 30.93 mAP even under the visual prompt configuration (Configuration **Conf 2**), and *Vehicle* drops from 32.12 mAP under the visual prompt configuration to 28.18 mAP under the multimodal configuration with oracle filtering (Configuration **Conf 5**). Mechanistically, general-domain ground imagery lacks prominent, top-down projected shadows. In overhead remote sensing, these dark shadows alter the object's top-down visual feature vector, pulling it away from the expected prompt representations in the shared feature space and inducing

TABLE III: Exhaustive Class-Wise Object Detection Evaluation on the DIOR Dataset. Metrics are reported for mAP (IoU 0.50:0.95) and AP@50. Modality Legend: T = Text-Only (**Conf 1**); TB = Text + Box (**Conf 3**); B = Box-Only (**Conf 2**); TF = Text + Oracle (**Conf 4**); TBF = Text + Box + Oracle (**Conf 5**).

	mAP (IoU 0.50:0.95)					**AP@50**				
Class Concept	**T**	**TB**	**B**	**TBF**	**TF**	**T**	**TB**	**B**	**TBF**	**TF**
Airplane	60.22	63.56	65.68	65.73	62.39	91.43	91.52	93.64	95.66	95.61
Airport	0	92.25	92.34	92.39	0	0	98.69	98.87	98.99	0
Baseball Field	0.01	59.87	77.54	60.98	0.01	0.01	66.14	89.63	67.54	0.01
Basketball Court	3.05	68.91	70.38	69.84	4.75	4.48	74.59	75.81	75.75	6.92
Bridge	1.32	39.57	39.37	39.97	1.58	3.14	51.32	50.73	52.04	3.81
Chimney	0.80	73.06	73.85	73.10	0.82	0.99	85.90	87.16	85.97	0.99
Dam	2.79	82.30	82.92	82.30	3.64	4.56	93.07	93.85	93.07	6.08
Expressway Service Area	0.91	67.75	70.94	68.60	2.20	1.03	82.36	87.98	83.61	2.58
Expressway Toll Station	2.92	82.24	83.97	83.17	8.37	4.92	91.40	92.18	92.80	13.53
Golf Field	0	84.31	87.31	86.02	0.35	0	89.74	93.87	91.90	0.50
Ground Track Field	0.63	73.27	78.24	74.22	1.78	1.20	77.33	83.59	78.72	3.45
Harbor	18.82	51.74	49.95	52.22	21.24	27.56	62.96	62.99	63.70	31.27
Overpass	0.52	48.80	50.80	50.88	2.02	1.21	62.31	65.04	65.87	4.64
Ship	14.45	34.52	44.23	34.56	14.49	21.89	51.35	67.30	51.41	21.96
Stadium	44.69	91.19	91.39	91.46	49.82	55.70	94.46	94.63	94.83	62.42
Storage Tank	25.10	46.92	46.34	47.22	25.88	38.20	69.53	67.18	70.09	39.46
Tennis Court	38.33	77.03	78.43	77.32	40.10	45.47	88.28	89.60	88.70	47.61
Train Station	0.63	85.47	84.39	85.47	0.63	0.99	97.91	97.31	97.91	0.99
Vehicle	17.55	24.49	32.12	28.18	21.47	32.00	38.15	53.42	46.78	41.41
Windmill	0.74	28.52	30.93	28.53	0.74	0.99	38.83	45.51	38.84	0.99

TABLE IV: Macro Performance Summary Across Prompting Setups on DIOR

Setup	**Overall mAP**	**Overall AP@50**
Text-Only (**Conf 1**)	11.67	16.79
Box-Only (**Conf 2**)	**66.55**	**79.51**
Text + Box (**Conf 3**)	63.79	75.29
Text + Oracle (**Conf 4**)	13.11	19.21
Text + Box + Oracle (**Conf 5**)	64.61	76.71

heavy coordinate regression jitter.

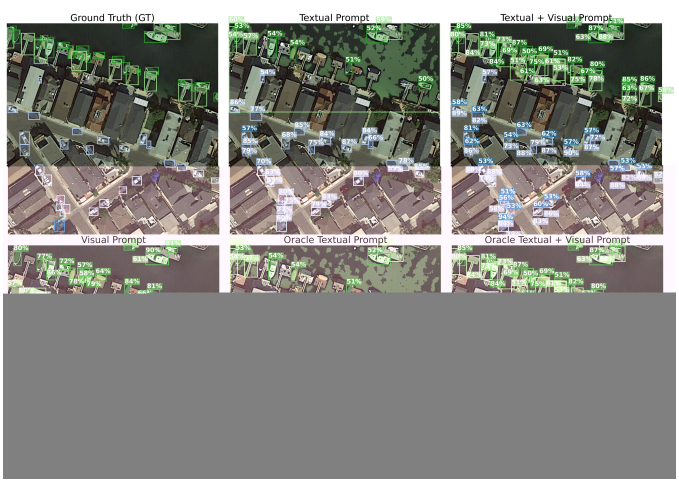
4) GZSD Benchmark Comparison and the Novel Split Bias: Using the Harmonic Mean proxy, we benchmark SAM 3 against Generalized Zero-Shot Detection (GZSD) architectures from the literature. Because the original evaluations of several foundational models did not include the DIOR dataset, the baseline metrics reported in Table V are sourced from the comparative evaluations conducted in CoseDet [16], VK-Det [21], and DescReg [22].

The benchmarking data reveals an important paradox regarding how dataset splits influence perceived generalizability. When evaluating raw visual localization, SAM 3 is highly dominant: its oracle Base AP_{50} (77.62) completely outclasses every single supervised, domain-adapted model in the literature, surpassing both Grounding DINO (70.80) [15] and the state-of-the-art CoseDet (68.10) [16]. This empirically verifies that SAM 3's physical feature representations are robustly world-class.

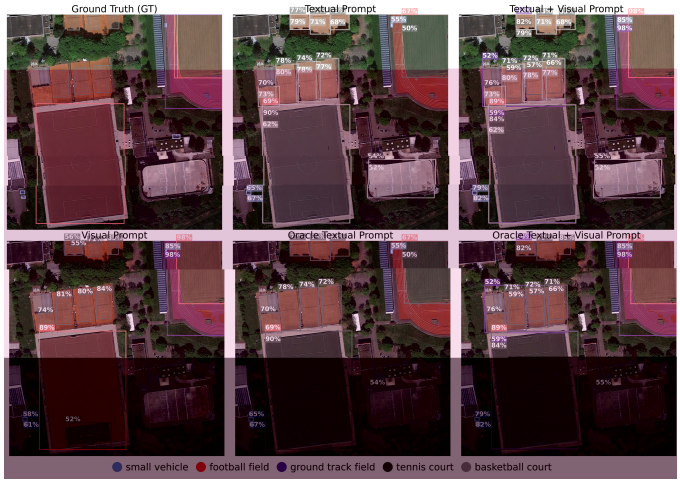
However, its final Harmonic Mean collapses severely to 5.48 (oracle), heavily dragged down by a catastrophic drop to 2.84 novel AP_{50}. This drop is largely an artifact of the specific classes the remote sensing community selected as "Novel" for the DIOR split (*Windmill, Airport, Baseball Field, Ground Track Field*). These categories represent sprawling, heterogeneous semantic zones rather than discrete geometric shapes. As shown in III, SAM 3's zero-shot text-only fails completely on these ambiguous boundaries, whereas it performs remarkably well on distinct geometric novel targets like airplanes (60.22 mAP) and tennis courts (38.33 mAP). Because the Harmonic Mean mathematically bottlenecks over the lowest constituent profile, this arbitrary split configuration obscures SAM 3's otherwise dominant general object detection capabilities.

TABLE V: GZSD Performance Comparison on the DIOR Dataset. Baseline performance metrics for foundational architectures are reported as reproduced by CoseDet [16], VK-Det [21], and DescReg [22].

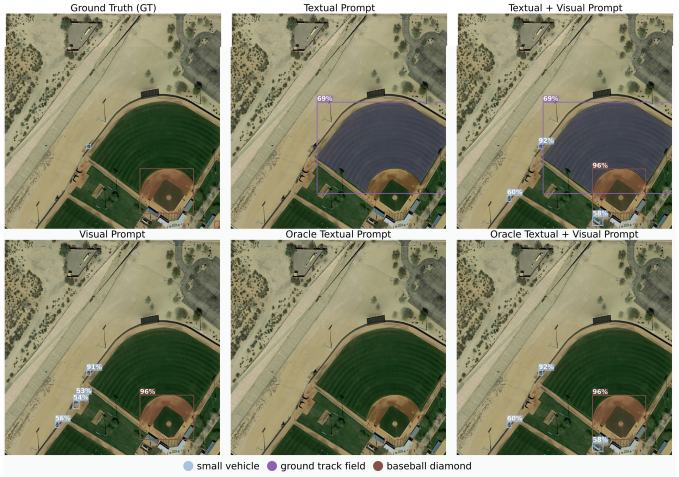
Model	**Avg mAP**	**Base** AP_{50}	**Novel** AP_{50}	**Harmonic Mean**
ViLD [23]	45.70	53.50	14.20	22.40
Grounding DINO [15]	57.30	70.80	3.20	6.20
YOLO-World [24]	57.70	70.20	8.00	14.40
CastDet [25]	58.80	62.70	43.40	51.30
CoseDet [16]	63.70	68.10	**46.10**	**53.50**
RRFS [14]	38.10	41.90	2.80	5.20
ContrastZSD [26]	41.90	51.40	3.90	7.20
DescReg [22]	—	68.70	7.90	14.20
VK-Det [21]	57.50	64.40	30.10	41.00
V2S [27]	—	57.00	1.40	2.70
SAM 3	61.15	76.03	1.67	3.26
SAM 3 Oracle	**62.66**	**77.62**	2.84	5.48



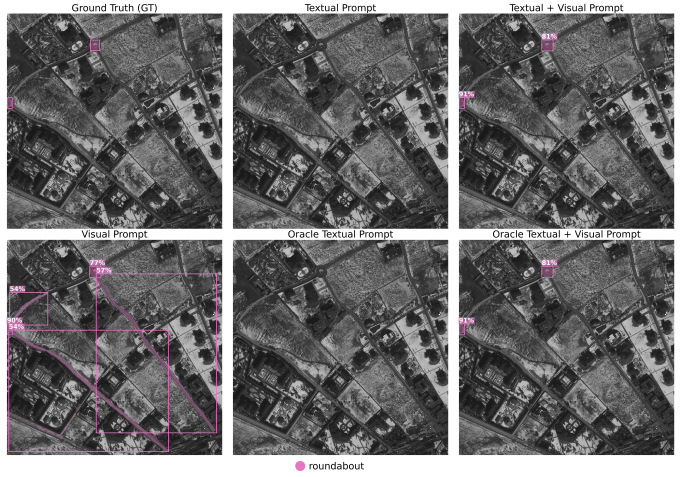
(a) Coastal Infrastructure



(b) Sports Fields



(c) Baseball Diamond



(d) Roundabout

Fig. 4: Qualitative instance segmentation results on the iSAID dataset across the prompt configurations (**Conf 1** through **Conf 5**). The multi-panel comparison demonstrates how spatial scale, visual ambiguity, and textual perspective gaps dictate the success or failure of the multimodal decoder.

C. Open-Vocabulary Instance Segmentation (iSAID)

Finally, we evaluate SAM 3's Promptable Concept Segmentation engine on the 15-class iSAID dataset. The macro performance profiles across the five functional modalities are summarized in Table VI. To mechanistically contextualize these quantitative metrics, a qualitative prompt ablation across four distinct spatial environments is presented in Figure 4.

TABLE VI: Macro Performance Summary Across Prompting Setups on iSAID

Setup	Overall mAP	Overall AP@50
Text-Only (**Conf 1**)	11.85	22.73
Box-Only (**Conf 2**)	32.77	57.20
Text + Box (**Conf 3**)	35.54	61.65
Text + Oracle (**Conf 4**)	12.70	24.15
Text + Box + Oracle (**Conf 5**)	**36.30**	**63.09**

1) Zero-Shot Baseline Deficit vs. One-Shot Synergistic Calibration: Ablating the prompting modalities on the iSAID dataset reveals a severe baseline performance deficit when relying strictly on textual guidance, with the Text-Only Configuration (**Conf 1**) bottlenecking at a low 11.85 mAP and 22.73 AP@50. Observing the visual outputs, the model completely fails to segment targets like roundabouts (Figure 4d) and baseball diamonds (Figure 4c) when given only a text prompt.

This total absence of detection empirically confirms a profound cross-modal domain gap within SAM 3's shared feature space. The text prompt generates a textual feature vector representing the semantic meaning of a "roundabout" or "baseball diamond." However, because the model associates these words with side-view imagery, their textual feature vectors sit far away from the top-view visual representations of these objects in the remote sensing image. In the case of the baseball diamond, the word "diamond" shifts the semantic meaning even further away from the actual concept. Because these representations are not close in the latent space, the Multimodal Decoder cannot find a match, resulting in zero segmentations.

However, fusing textual and visual modalities (**Conf 3**) for

instance segmentation actively improved overall accuracy to 35.54 mAP and 61.65 AP@50. Observing the roundabout in Figure 4d, a purely visual prompt causes the model to segment the surrounding roads. Similarly, for the harbor in Figure 4a, the visual prompt alone segments the nearby houses, while the textual prompt alone mistakenly segments the water. When both text and visual prompts are mixed, the model correctly segments the roundabout and perfectly outlines the harbor docks.

This improvement demonstrates synergistic cross-modal calibration within the feature space. The visual feature representation of a harbor is very similar to a building because they share the same materials and colors. Meanwhile, the textual feature vector of a harbor is more closely associated with water. When these prompts are mixed, the textual and visual representation vectors eliminate each other's ambiguity. The visual vector prevents the model from selecting water, and the textual vector prevents it from selecting buildings, shifting the combined representation perfectly toward the visual representation of the harbor in the image. The same occurs for the roundabout, where the textual prompt helps pull the visual representation away from standard roads.

Yet, this mixed prompting can also introduce interference, as observed with the vehicles in Figure 4a. When queried with only the textual prompts for "small vehicle" and "large vehicle," the model is better able to distinguish between the two concepts. However, when queried with visual prompts, or a mix of visual and textual prompts, the model segments both small and large vehicles indiscriminately.

Mechanistically, in the textual feature space, the words "small" and "large" are further apart, allowing the model to understand the semantic difference between the two concepts. Conversely, the top-down visual feature representations of small and large vehicles are almost identical. When mixing the visual and textual prompts, the feature vectors of the words "small" or "large" are interpreted as noise. The Multimodal Decoder focuses entirely on the stronger, shared visual representation of the concept "vehicle," losing the ability to distinguish the specific sub-classes.

2) Presence Head Hallucinations and Filtering: Implementing presence query filtering (Configurations **Conf 4** and **Conf 5**) directly validates the vulnerability of the decoupled presence head. The unfiltered pipelines suffer from clear semantic hallucinations, mistakenly forcing background features into targeted masks.

Observing the visual outputs, when the model is prompted with the full vocabulary of iSAID concepts, it hallucinates incorrect classes over distinct visual features. In Figure 4b, the model mistakenly segments the tennis courts as basketball courts. Similarly, in Figure 4c, the model incorrectly detects the generic green grass of the baseball outfield as a ground track field.

Mechanistically, these errors are driven by textual-visual misalignment in the shared feature space. The top-down visual feature vectors of rectangular sports courts (tennis and basketball) are highly similar. When the textual feature vector for "basketball court" is introduced, the model cannot distinguish the subtle geometric differences from above, leading to semantic confusion. Likewise, the textual feature vector for a "ground track field" becomes overly associated with the visual feature vector of generic green terrain, causing the model to hallucinate a track where only grass exists.

Applying the oracle negative filters actively suppresses this background noise. As seen in the oracle configurations of Figures 4b and 4c, the false-positive masks for the basketball courts and ground track fields are completely removed. This filtering lifts the zero-shot Text-Only mode to 12.70 AP and 24.15 AP@50 and the multimodal setup to 36.30 mAP and 63.09 AP@50. However, the relatively small margin of improvement ($\sim$ 1.5 to 2 points) mathematically confirms that while the presence head introduces false-positive hallucinations, the primary operational bottleneck remains false negatives driven by the feature-space domain gap.

3) Granular Class Analysis: Persistent Vulnerabilities and Structural Drops: A rigorous, class-by-class examination of the exhaustive data grid (Table VII) strips away the statistical smoothing of macro averages, exposing severe, localized failures that persist across the sam 3 model:

- **Complete Semantic Deficits:** Observing the data for complex categories like *Baseball Diamond*, performance collapses to an absolute 0 mAP under the pure zero-shot text modality (Configuration **Conf 1**). As visually confirmed in Figure 4c, the model simply returns empty predictions. Mechanistically, the textual feature vector of a baseball diamond is so far removed from its top-down visual feature vector that the Multimodal Decoder cannot make a connection. While adding a visual prompt (Configuration **Conf 5**) successfully anchors the target in the feature space and elevates the score to 71.03 AP@50, the zero-shot failure highlights an acute semantic blind spot.
- **Micro-Scale and Clutter Vulnerabilities:** A critical structural weakness is observed in the *Small Vehicle* category, which scores a severely depressed 6.76 mAP in zero-shot text mode and stalls heavily at 9.15 mAP under optimal multimodal filtering. Observing Figure 4a, the model constantly conflates small and large vehicles regardless of the prompt. This is fundamentally a resolution and feature-space limitation. Densely packed, sub-pixel targets lose distinctive visual details during the Image Encoder's pooling stages. Consequently, the extracted visual feature vectors for small and large vehicles become nearly identical in the latent space, making it impossible for the model to separate the specific sub-classes.
- **Amorphous Boundary Failures:** Visually complex, poorly bounded regions like *Harbor* display an acute resistance to high-precision segmentation, scoring a low 4.00 mAP in text mode and managing only 24.78 mAP under optimal multimodal conditions. As observed in Figure 4a, the visual boundaries of a harbor inherently blend into adjacent water and building structures. In the shared feature space, the visual feature vector of the harbor overlaps heavily with the visual vectors of these surrounding elements. While mixing modalities successfully shifts the core prediction toward the docks, this overlapping visual similarity continuously induces

TABLE VII: Class-Wise Instance Segmentation Evaluation on the iSAID Dataset. Metrics are reported for mAP (IoU 0.50:0.95) and AP@50. Modality Legend: T = Text-Only (**Conf 1**); TB = Text + Box (**Conf 3**); B = Box-Only (**Conf 2**); TF = Text + Oracle (**Conf 4**); TBF = Text + Box + Oracle (**Conf 5**).

	mAP (IoU 0.50:0.95)					**AP@50**				
Class Concept	**T**	**TB**	**B**	**TBF**	**TF**	**T**	**TB**	**B**	**TBF**	**TF**
Large Vehicle	26.84	31.14	28.68	31.79	28.04	47.13	57.11	52.96	58.53	49.60
Baseball Diamond	0	41.48	43.03	41.48	0	0	71.03	78.06	71.03	0
Basketball Court	0.25	44.90	44.49	50.79	1.40	0.61	66.03	67.34	75.81	3.30
Bridge	0.52	18.18	18.12	19.61	1.40	1.16	44.12	44.45	48.64	3.49
Football Field	20.40	64.00	46.91	64.14	21.88	27.57	83.04	63.15	83.27	29.54
Ground Track Field	0.01	23.29	24.61	24.82	0.09	0.02	39.53	41.76	42.33	0.27
Harbor	4.00	24.73	25.69	24.78	4.18	13.15	60.02	60.29	60.16	13.64
Helicopter	2.26	12.18	7.29	12.18	2.27	13.16	46.68	25.51	46.68	13.19
Plane	29.60	33.20	30.70	33.20	29.70	80.98	83.68	77.14	83.86	81.21
Roundabout	8.14	64.23	54.71	64.52	11.72	12.84	86.15	73.75	86.64	18.42
Ship	8.35	26.80	27.70	26.90	8.50	14.05	51.93	52.94	52.11	14.30
Small Vehicle	6.76	9.11	8.05	9.15	6.96	18.33	24.62	20.65	24.75	18.60
Storage Tank	19.67	39.62	36.00	39.83	20.42	30.73	66.36	61.38	66.83	31.94
Swimming Pool	18.28	29.91	29.23	29.93	18.31	42.37	61.17	59.67	61.22	42.45
Tennis Court	32.70	70.42	65.40	71.36	35.59	38.86	83.22	78.98	84.50	42.33

TABLE VIII: GZSI Performance Comparison on the iSAID Dataset. Baseline performance metrics for architectures are reported as reproduced by ZoRI [17].

Model	**Base** AP_{50}	**Novel** AP_{50}	**Harmonic Mean**
ZSI [6]	45.83	0.64	1.26
D2Zero [5]	39.80	0.39	0.77
FC-CLIP [28]	43.47	4.91	8.83
ZoRI (AAAI 2025) [17]	47.05	9.30	15.53
SAM 3	59.14	30.49	40.24
SAM 3 Oracle	**60.97**	**31.88**	**41.87**

boundary confusion at the pixel level that neither text nor a single visual exemplar can completely resolve.

4) GZSI Benchmark Evaluation and Objective Assessment: To project these capabilities into standard comparative literature, we executed our proxy evaluation protocol against the Generalized Zero-Shot Instance Segmentation (GZSI) standard. To ensure a standardized comparative baseline, the performance metrics for legacy and foundational GZSI architectures reported in Table VIII are sourced directly from the comparative evaluations conducted by the state-of-the-art ZoRI framework [17].

The comparative data in Table VIII highlights a critical trade-off in current methodologies. Traditional closed-vocabulary models (ZSI [6], D2Zero [5], FC-CLIP [28]) exhibit a strong inductive bias toward their training distributions, resulting in severe performance degradation on novel classes (e.g., ZSI drops to 0.64 AP@50). By operating as a frozen foundation model, SAM-3 intrinsically bypasses this closed-set overfitting. In this specific iSAID split, the community-designated unseen categories (*helicopter, football field, swimming pool, tennis court*) possess distinct, rigid geometric layouts that align well with the spatial priors learned during SAM-3's general-domain pre-training. Consequently, our unfiltered proxy achieves a 30.49 Unseen AP@50, yielding a balanced performance profile and a Harmonic Mean of 40.24 (41.87 filtered).

However, this elevated Harmonic Mean must be interpreted objectively. While SAM-3 outperforms specialized GZSI architectures purely by avoiding novel-class degradation, its absolute precision remains constrained. The model's low overall mAP, combined with systemic vulnerabilities regarding small-scale targets and ambiguous boundaries, demonstrates that an untuned foundation model is not yet sufficient for automated, operational GIS pipelines.

Ultimately, these experiments delineate the operational boundaries of SAM-3 in the remote sensing domain: the architecture possesses a highly robust spatial feature extraction backbone, yet it is severely restricted by cross-modal semantic misalignment and finite resolution thresholds. This underscores a clear trajectory for future research: domain-specific fine-tuning is strictly required not only to calibrate its textual projection layers, but also to optimize its visual encoders for the high-density, small-scale targets typical of Earth Observation tasks.

V. Conclusion and Future Work

This study presents one of the first comprehensive, multi-task evaluations of the Segment Anything Model 3 (SAM 3) in the Earth Observation domain. Driven by the stark performance drop SAM 3 exhibits on highly specialized datasets, we systematically benchmarked the model across scene classification (AID), open-vocabulary object detection (DIOR), and instance segmentation (iSAID) without any domain-specific fine-tuning. By mapping the operational boundaries of its Promptable Concept Segmentation (PCS) architecture, our findings reveal that SAM 3 has a promising potential for remote sensing applications: it possesses a world-class spatial engine capable of unrivaled zero-shot localization, yet it

remains severely bottlenecked by a profound semantic domain gap.

The core scientific discovery of this evaluation lies in diagnosing exactly where this failure occurs: the multimodal decoder. Architecturally, SAM 3 integrates three distinct latent representations–the whole-image feature vector, the visual prompt feature vector, and the textual prompt feature vector. Our 5-modality ablation studies demonstrate that when relying exclusively on visual prompts (one-shot exemplars), the multimodal decoder perfectly aligns the search query to the complex remote sensing geometry, allowing SAM 3 to consistently outperform state-of-the-art, domain-adapted supervised models. However, when a textual prompt vector is introduced, performance collapses. Because the model's text-to-vision alignment was pre-trained exclusively on ground-level internet imagery, the textual prompt vector injects misaligned, egocentric semantic expectations into the decoder's cross-attention layers, leading to the severe negative cross-modal interference observed in our detection tasks.

These findings chart a highly specific and promising trajectory for future research. Because SAM 3's massive Vision Transformer backbone perfectly parses the dense, complex geometry of aerial imagery out-of-the-box, the remote sensing community does not need to expend massive computational resources to retrain its visual cortex. Instead, future work should focus entirely on lightweight, Parameter-Efficient Fine-Tuning (PEFT) techniques–such as Low-Rank Adaptation (LoRA) or vision-language contrastive realignment–targeting only the text encoder and the multimodal decoder. By training the model to align remote sensing vocabulary directly with top-down visual embeddings, researchers can eliminate the cross-modal interference without degrading the foundational visual engine. Resolving this alignment gap will unlock SAM 3's full potential, transforming it from a ground-level generalist into a definitive, unified foundation model for automated geospatial intelligence.

REFERENCES

[1] A. Kirillov, E. Mintun, N. Ravi *et al.*, "Segment anything," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 4015–4026.

[2] N. Ravi, V. Gabeur, Y.-T. Hu *et al.*, "Sam 2: Segment anything in images and videos," *arXiv preprint arXiv:2408.00714*, 2024.

[3] N. Carion, L. Gustafson, Y.-T. Hu *et al.*, "Sam 3: Segment anything with concepts," *arXiv preprint arXiv:2511.16719*, 2026.

[4] F. Liu, D. Chen, Z. Guan *et al.*, "Remoteclip: A vision language foundation model for remote sensing," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.

[5] S. He, H. Ding, and W. Jiang, "Semantic-promoted debiasing and background disambiguation for zero-shot instance segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[6] Y. Zheng, J. Wu, Y. Qin *et al.*, "Zero-shot instance segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 2593–2602.

[7] G.-S. Xia, J. Hu, F. Hu *et al.*, "Aid: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 7, pp. 3965–3981, 2017.

[8] K. Li, G. Wan, G. Cheng *et al.*, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 159, pp. 296–307, 2020.

[9] S. W. Zamir, A. Arora, A. Gupta *et al.*, "isaid: A large-scale dataset for instance segmentation in aerial images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2019, pp. 28–37.

[10] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*. PMLR, 2021, pp. 8748–8763.

[11] Z. Zhang, T. Zhao, Y. Guo *et al.*, "Rs5m and georsclip: A large-scale vision-language dataset and a large vision-language model for remote sensing," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.

[12] A. Terlizzi, A. Nazzaro, L. Bernardi *et al.*, "Rsdix: Lightweight and data-efficient vlms for remote sensing through self-distillation," *International Joint Conference on Neural Networks (IJCNN)*, 2025.

[13] N. Carion, F. Massa, G. Synnaeve *et al.*, "End-to-end object detection with transformers," in *European conference on computer vision*. Springer, 2020, pp. 213–229.

[14] P. Huang, J. Han, D. Cheng *et al.*, "Robust region feature synthesizer for zero-shot object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 7622–7631.

[15] S. Liu, Z. Zeng, T. Ren *et al.*, "Grounding dino: Marrying dino with grounded pre-training for open-set object detection," in *European Conference on Computer Vision (ECCV)*. Springer, 2024.

[16] J. Yao, Y. Zheng, S. Lu *et al.*, "Cosedet: Open-vocabulary remote sensing object detection with contextual semantic information," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 18, pp. 1120–1135, 2025.

[17] S. Huang, H. He, and W. Ding, "Zori: Towards discriminative zero-shot remote sensing instance segmentation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[18] Z. Wang, R. Prabha, T. Huang *et al.*, "Skyscript: A large and semantically diverse vision-language dataset for remote sensing," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 5805–5813, 2024.

[19] X. Li, C. Wen, Y. Hu *et al.*, "Rs-clip: Zero shot remote sensing scene classification via contrastive vision-language supervision," *International Journal of Applied Earth Observation and Geoinformation*, vol. 124, p. 103497, 2023.

[20] Y. He *et al.*, "Enhancing remote sensing vision-language models through mllm and llm-based high-quality image-text dataset generation," *arXiv preprint arXiv:2507.16716*, 2025.

[21] J. Yao, Y. Zheng, S. Lu *et al.*, "Vk-det: Visual knowledge guided prototype learning for open-vocabulary aerial object detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026.

[22] X. Zhu, J. Pang, and C. Wang, "Zero-shot object detection via descriptor registration," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[23] X. Gu, T.-Y. Lin, W. Kuo *et al.*, "Open-vocabulary object detection via vision and language knowledge distillation," *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.

[24] B. Cheng, I. Misra, A. G. Schwing *et al.*, "Yolo-world: Real-time open-vocabulary object detection," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[25] Y. Li, W. Guo, X. Yang *et al.*, "Castdet: Toward open vocabulary aerial object detection with clip-activated student-teacher learning," *European Conference on Computer Vision (ECCV)*, 2024.

[26] C. Yan, X. Chang, M. Luo *et al.*, "Semantics-guided contrastive network for zero-shot object detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 3, pp. 1530–1547, 2024.

[27] S. Khandelwal, A. Nambirajan, B. Siddiquie *et al.*, "Frustratingly simple but effective zero-shot detection and segmentation: Analysis and a strong baseline," *arXiv preprint arXiv:2302.07319*, 2023.

[28] Q. Yu, J. He, X. Deng *et al.*, "Convolutions die hard: Open-vocabulary segmentation with single frozen convolutional clip," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.